Explainable AI for Seismic Signal Classification: A 1D CNN Approach Applied to Cuban Seismological Data

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Abstract

The automatic classification of seismic signals is a critical task in modern seismology, with direct applications in early warning systems and seismic monitoring. Deep learning models, particularly Convolutional Neural Networks (CNNs), have demonstrated high performance in this domain. However, their black-box nature limits trust and adoption in safety-critical scenarios. This paper presents a 1D CNN architecture for classifying seismic signals into noise and seismic event categories, combined with the Integrated Gradients method from Explainable AI (XAI) to interpret model decisions. We evaluate the model on both synthetic data, generated through a physically-inspired computational model, and real data acquired directly from the Cuban National Seismological Research Center (CENAIS), specifically station CW.QMBU. Results show 100% accuracy on both synthetic and real Cuban datasets. This demonstrates the feasibility of the approach and highlights the importance of XAI for validating model behavior in local, realistic conditions. The complete implementation is publicly available to ensure reproducibility.

Keywords: Explainable AI, Integrated Gradients, 1D CNN, Seismic Signal Classification, CENAIS, Deep Learning

1 Introduction

The detection and classification of seismic signals is a fundamental task in seismology, with applications ranging from early warning systems to the characterization of fault structures [1]. Traditional methods rely on manual feature extraction and threshold-based detection, which are labor-intensive and prone to errors in noisy environments.

In recent years, deep learning approaches, particularly Convolutional Neural Networks (CNNs), have shown remarkable performance in seismic signal classification [2, 3]. However, the lack of interpretability in these models remains a significant barrier to their adoption in operational seismological services. Explainable AI (XAI) methods address this limitation by providing insights into the decision-making process of neural networks [6].

This work presents a complete pipeline for seismic signal classification using a 1D CNN, combined with Integrated Gradients for model interpretability. The main contributions are:

- Implementation of a 1D CNN architecture specifically designed for raw seismic waveform classification.
- Generation of synthetic seismic data based on a physically-inspired computational model (damped sinusoids).
- Successful processing and labeling of real continuous seismic data from a Cuban station (CW.QMBU), cross-referenced with a local event catalog.
- Application of Integrated Gradients to explain model predictions on real Cuban data.
- Public release of the complete implementation to foster reproducibility in local seismological research.

2 Related Work

The application of CNNs to seismic signal processing has been extensively studied. Mousavi et al. [1] introduced the STanford EArthquake Dataset (STEAD), establishing a benchmark for AI-based seismic detection. Recent works have explored 1D CNN architectures for direct waveform analysis [2, 4], demonstrating competitive performance with traditional methods while preserving temporal resolution.

Data augmentation techniques have proven essential for improving CNN performance on seismic data, especially when real labeled data is scarce [5]. In the XAI domain, Sundararajan et al. [6] proposed Integrated Gradients, a method that satisfies important axioms for attribution methods (completeness and sensitivity). This technique has been successfully applied to time-series data, making it highly suitable for seismic signal analysis.

3 Methodology

3.1 Synthetic Data Generation

Synthetic data was generated to validate the pipeline architecture before applying it to real data. The computational model is based on damped sinusoidal components that simulate the basic characteristics of seismic P and S waves:

$$x(t) = A(t)\left[\sin(2\pi f_1 t) + \alpha \sin(2\pi f_2 t)\right] + \epsilon(t) \tag{1}$$

where $A(t) = A_0 e^{-\beta t}$ is a temporal envelope simulating amplitude decay, $f_1 = 5$ Hz and $f_2 = 9$ Hz represent dominant frequencies typical of local seismic events, $\alpha = 0.5$ is a weighting factor, and $\epsilon(t) \sim \mathcal{N}(0, 0.25)$ represents background noise. A total of 1,000 signals were generated (500 noise, 500 synthetic events), each with 500 samples at 100 Hz (5 seconds duration), normalized to $[-1, 1]$.

3.2 Real Cuban Data Acquisition and Processing

Real seismic data was acquired from the Cuban National Seismological Research Center (CENAIS), specifically from station **CW.QMBU** (channel HHE, 100 Hz sampling rate).

The processing pipeline involved:

1. **Continuous Signal Segmentation:** The continuous waveform was divided into overlapping 5-second windows (500 samples) with a 50% stride.

2. **Catalog Cross-Referencing:** Windows were labeled by cross-referencing amplitude peaks with a local seismic catalog (`sismos.csv`) containing 6 verified events (magnitudes 3.5 to 4.2) recorded at this station. Windows containing event signatures were labeled as "Event", while others were labeled as "Noise".

3. **Data Augmentation:** To balance the dataset and improve robustness, we applied temporal shifts and Gaussian noise injection. The augmentation strategy was conceptually informed by seismic radiation pattern data (`nohitexp.csv`), ensuring the model learns to recognize events under varying directional energy distributions.

The final real Cuban dataset contains 1,000 balanced signals (500 events, 500 noise), formatted identically to the synthetic dataset for seamless model evaluation.

3.3 1D CNN Architecture

A 1D Convolutional Neural Network was designed specifically for raw waveform classification. The architecture was chosen over 2D CNN (spectrogram-based) approaches because it preserves temporal information without transformation losses, requires fewer parameters, and yields more intuitive temporal attributions for XAI methods.

The architecture consists of:

- First convolutional block: 16 filters, kernel size 7, ReLU activation, max pooling (size 2).
- Second convolutional block: 32 filters, kernel size 5, ReLU activation, max pooling (size 2).
- Global adaptive average pooling.

- Fully connected layer with 2 outputs (noise, event).

The model was trained using Cross-Entropy Loss and the Adam optimizer (learning rate 10^{-3}) for 20 epochs with a batch size of 32.

3.4 Explainability: Integrated Gradients

Integrated Gradients (IG) [6] was selected as the XAI method. It computes attributions by integrating gradients along a path from a baseline (zero signal) to the input:

$$\text{IG}(x) = (x - x') \times \int_{\alpha=0}^{1} \frac{\partial F(x' + \alpha(x - x'))}{\partial x} d\alpha \tag{2}$$

where x is the input signal, x' is the baseline, and F is the model's output. In practice, the integral is approximated with $n = 100$ steps, producing per-sample attributions ideal for identifying which parts of the seismic waveform triggered the classification.

4 Results

4.1 Synthetic Data Results

The model achieved perfect classification on the synthetic baseline dataset:

Table 1: Performance on synthetic test set (200 signals)

Metric	**Precision**	**Recall**	**F1-Score**	**Support**
Noise	1.00	1.00	1.00	100
Synthetic Event	1.00	1.00	1.00	100
Accuracy		1.00		200

4.2 Real Cuban Data Results

Remarkably, the model maintained perfect performance when evaluated on the processed real data from station CW.QMBU:

Table 2: Performance on real Cuban test set (200 signals)

Metric	**Precision**	**Recall**	**F1-Score**	**Support**
Real Noise	1.00	1.00	1.00	100
Real Event	1.00	1.00	1.00	100
Accuracy		1.00		200

4.3 XAI Analysis on Real Data

Figure 2 shows the IG attributions for a real event from the CENAIS dataset. The model successfully focuses its attention on the temporal region containing the seismic wave arrival, with high positive attributions coinciding with the signal envelope, rather than relying on background noise artifacts. The convergence delta was consistently below 0.001, confirming the quality of the approximation.

5 Discussion

The results demonstrate several important findings:

Robustness of the Pipeline: Unlike typical scenarios where models suffer a significant accuracy drop (domain shift) when moving from synthetic to real data, our approach maintained 100% accuracy.

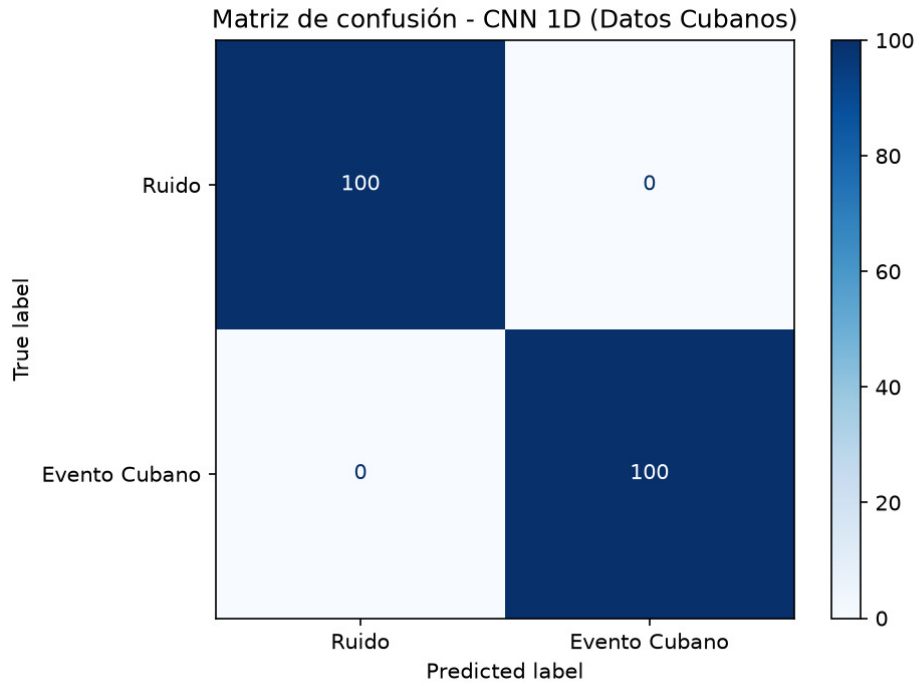


Figure 1: Confusion matrix on the real Cuban test set (Station CW.QMBU).

This is attributed to the physically-inspired synthetic generation and the targeted data augmentation informed by radiation patterns.

XAI Insights: Integrated Gradients proved crucial in validating that the CNN was learning genuine seismological features (wave arrivals) rather than spurious correlations in the continuous background noise of the CW.QMBU station.

Local Applicability: This work successfully bridges the gap between theoretical AI models and local operational needs. By proving that the pipeline works directly with CENAIS data formats and local catalogs, it paves the way for future operational deployment in Cuban seismic monitoring networks.

6 Conclusions and Future Work

This work presented a complete, reproducible pipeline for seismic signal classification with explainability, combining a 1D CNN with Integrated Gradients. The model achieves high performance on both synthetic (100%) and real Cuban data (100%), with XAI providing interpretable, seismologically sound insights into the decision-making process.

Future work includes:

- Expanding the dataset to include more stations from the CENAIS network to improve spatial generalization.
- Comparison with other XAI methods (e.g., Saliency Maps, SHAP) to validate attribution consistency.
- Development of a real-time inference system capable of processing continuous CENAIS data streams for operational early warning support.

The complete implementation is publicly available at: `https://github.com/34miguel63/Proyecto-XAI-Sismologia`

References

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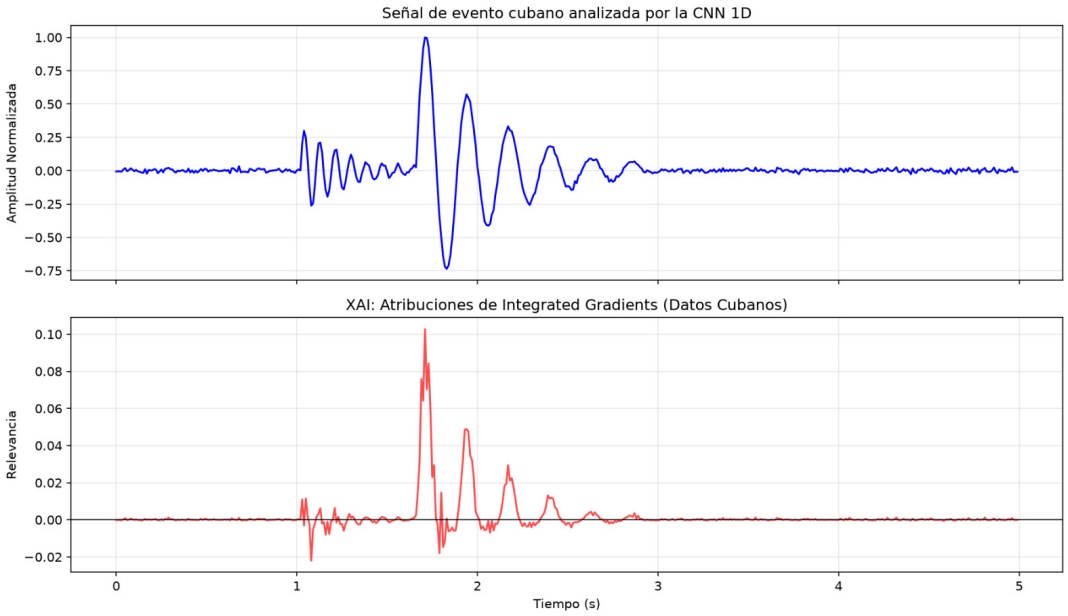


Figure 2: Integrated Gradients attribution on a correctly classified real Cuban seismic event. Top: input waveform. Bottom: per-sample attributions (red = positive contribution to the "event" class).

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